Protocol S1

Primer sequences

Human loci ApoB, D18S51, Sry and PBDG:

apoB4	CAGTGTATCTGGAAAGCCTACAGGACACCAAAA
apoB300	TGCTTTCATACGTTTAGCCCAATCTTGGATAG
apoB700	TGGTAAACGGAAGTCTGGCAGGGTGATTCTCG
apoB800	CAATTGTGTGTGAGATGTGGGGAAGCTGGAAT
apoB900	GAGGTGGTTCCATTCCCTATGTCAGCATTTGC
apoB1000	GGGTTTGAGAGTTGTGCATTTGCTTGAAAATC
apoB500	ATGGTAAATTCTGGTGTGGAAAACCTGGATGG
apoB500-28	TAAATTCTGGTGTGGAAAACCTGGATGG
apoB500-25	ATTCTGGTGTGGAAAACCTGGATGG
apoB300rev	CTATCCAAGATTGGGCTAAACGTATGAAAGCA
apoB300rev-28	CCAAGATTGGGCTAAACGTATGAAAGCA
apoB300rev-25	AGATTGGGCTAAACGTATGAAAGCA
D18S51 5'-28	GACATGTTGGCTTCTCTCTGTTCTTAAC
D18S51 5'-25	ATGTTGGCTTCTCTCTGTTCTTAAC
D18S51 3'-28	GCACGTGCCTGTAGTCTCAGCTACTTGC
D18S51 3'-25	CGTGCCTGTAGTCTCAGCTACTTGC
sry3	AAAGCTGTAACTCTAAGTATCAGTGTGAAAC
sry3-28	GCTGTAACTCTAAGTATCAGTGTGAAAC

sry3-25	GTAACTCTAAGTATCAGTGTGAAAC
sry4	GTTGTCCAGTTGCACTTCGCTGCAGAGTACC
sry4-28	GTCCAGTTGCACTTCGCTGCAGAGTACC
sry4-25	CAGTTGCACTTCGCTGCAGAGTACC
pbdg5	AACAGCCCAAAGATGAGAGTGATTCGCGTGG
pbdg6	AGTCCTGCCCTTCCCAGCCACGGCTCAGAC

B. subtilis locus SpoB:

bs1	ACGGCATTAACAAACGAACTGATTCATCTGCTTGG
bs2	CCTTAATTTCTCCGAGAACTTCATATTCAAGCGTC

Sequences of MRSA/MSSA alleles

MRSA/MSSA primers:

sccI/II	CTCAAAGCTAGAACTTTGCTTCACTATAAGTATTC
sccIII	CCAATATTTCATATATGTAATTCCTCCACATCTCA
orfX	CCCAAGGGCAAAGCGACTTTGTATTCGTCATTGGCGGATCAAACG
mssa	CCAATTTGATAGGGCCTAATTTCAACTGTTAGCTA

Genomic target sequences (primer target sites are bold/underlined, probe binding sites are in bold/italic):

MSSA

TTTTAGATATAAA**CCAATTTGATAGGGCCTAATTTCAACTGTTAGCTA**CTA

CTTAAGTTATATGCGCAATTATCGTGATATATCTTATATATTGAATGAACGTG

GATTTAATGTCCACCATTTAACACCCTCCAAATTATTATCTCCTCATACAGAA

TTTTTTAGTTTTACTTATGATACGCCT***CTCCACGCATAATCTTAAATGCTCTATA***

CACTTGCTCAATTAACACAACCCGCATCATTTGATGTGGGAATGTCATTTTGCT

GAATGATAGTGCGTAGTTACTGCGTTGTAAGACGTCCTTGTGCAGGC**CGTTT**

GATCCGCCAATGACGAATACAAAGTCGCTTTGCCCTTGGGTCATGCG

MRSAI:

TTTAGTTGCAGAAAGAATTTT**CTCAAAGCTAGAACTTTGCTTCACTATAAG**

TATTCAGTATAAAGAATATTTCGCTATTATTTACTTGAAATGAAAGACTGCG

GAGGCTAACTATGTCAAAAATCATGAACCTCATTACTTATGATAAGCTT***CTCC***

ACGCATAATCTTAAATGCTCTATACACTTGCTCAATTAACACAACCCGCATCAT

TTGATGTGGGAATGTCATTTTGCTGAATGATAGTGCGTAGTTACTGCGTTGTA

AGACGTCCTTGTGCAGGC**CGTTTGATCCGCCAATGACGAATACAAAGTCG**

CTTTGCCCTTGGGTCATGCG

MRSAII:

TTTAGTTGCAGAAAGAATTTT**CTCAAAGCTAGAACTTTGCTTCACTATAAG**

TATTCAGTATAAAGAATATTTCGCTATTATTTACTTGAAATGAAAGACTGCG

GAGGCTAACTATGTCAAAAATCATGAACCTCATTACTTATGATAAGCTTCTTA

AAAACATAACAGCAATTCACATAAACCTCATATGTTCTGATACATTCAAAAT

CCCTTTATGAAGCGGCTGAAAAAACCGCATCATTTATGATATGCTT***CTCCACG***

CATAATCTTAAATGCTCTGTACACTTGTTCAATTAACACAACCCGCATCATTTG
ATGTGGGAATGTCATTTTGCTGAATGATAGTGCGTAGTTACTGCGTTGTAAGA
CGTCCTTGTGCAGGC**CGTTTGATCCGCCAATGACGAATACAAAGTCGCTTT**
GCCCTTGGGTCATGCG

MRSAIII:

AAGGTATAAT**CCAATATTTCATATATGTAATTCCTCCACATCTCA**TTAAAT
TTTTAAATTATACACAACCTAATTTTTAGTTTTATTTATGATACGCTT***CTCCAC***
GCATAATCTTAAATGCTCTGTACACTTGTTCAATTAACACAACCCGCATCATTT
GATGTGGGAATGTCATTTTGCTGAATGATAGTGCGTAGTTACTGCGTTGTAAG
ACGTCCTTGTGCAGGC**CGTTTGATCCGCCAATGACGAATACAAAGTCGCTT**
TGCCCTTGGGTCATGCG

RPA conditions

RPA relies on the establishment of a reaction environment that support the formation of recombinase-oligonucleotide complexes. Since the process is also ATP-dependent [3], it requires an energy regeneration system for sustained activity. We titrated key components of the RPA reaction mixture in order to determine their influence on amplification performance (Fig. S1). RPA proved to work robustly over a relatively wide range of reagent concentrations. We found, however, that optimal reaction conditions varied between different primer pairs and therefore had to be defined individually.

Primer requirements

We used RPA to amplify of a wide range of targets. While the design of primers revealed no limitations on sequence composition itself, certain parameters have to be met for an oligonucleotide to be suitable for RPA. Amplicon size under the conditions employed is limited to approximately 1kb (Fig. S2). The minimum length of a primer proved to be about 30 nucleotides (Fig. S3). We observed variability in the performance of oligonucleotides that differ in sequences but are similar in length and position relative to their counterpart. The rules governing the influence of nucleotide sequence on the quality of a particular RPA primer are currently under investigation.

Control DNA

The wild-type *S. aureus* DNA (MSSA) [11,12] serving as a negative control in the experiment shown in 2C does act as a template for RPA when combined with the primer pair orfX/mssa (Fig. S5).